Skills Summary

Ordering Fractions and Mixed Numbers

Use this reference while completing your worksheet or when studying.

Fractions can only be ordered once they are in *one* form. Rewrite first, compare second.

1. Equivalent forms: rewriting without changing value

Rule: multiplying (or dividing) the numerator *and* the denominator by the same non-zero number never changes the value.

$\frac{a}{b} = \frac{a \times k}{b \times k}$ find k from the denominators, then apply it to the numerator.

Mixed number → fraction: each whole is one full denominator, so multiply the whole part by the denominator and add the numerator.

Fraction → mixed number: divide; the quotient is the whole part and the remainder stays over the same denominator.

Example: Rewrite $\frac{2}{3}$ with denominator 15.
Step 1. The denominator is fixed for me, so find the scale factor from the denominators first and then apply the same factor upward.
Step 2. 3 scaled to 15 is a factor of 5, so the numerator scales the same way: $2 \times 5 = 10$.
$\frac{2}{3} = \mathbf{\frac{10}{15}}$

Try it: Rewrite $\frac{3}{5}$ with denominator 20.
check: $\mathbf{\frac{12}{20}}$

2. Comparing a pair: benchmark, then common denominator

Try these in order — stop at the first one that decides it:

- **Benchmark.** Is each number below a half, above a half, or above one? Half of a fraction's denominator, written as the numerator, is exactly one half.
- **Gap to one.** If both sit just below one, the one with the *smaller* missing piece is larger.
- **Matching tops or bottoms.** Same numerator: bigger denominator means smaller pieces, so a smaller number. Same denominator: compare the numerators.
- **Common denominator.** Always works. Multiply the two denominators if no smaller common one is obvious.

Example: Which is greater, $\frac{4}{9}$ or $\frac{5}{8}$?
Step 1. The two numbers look close, so test the half benchmark before doing any arithmetic — it is the cheapest test that can settle it.
Step 2. Half of 9 would be a numerator between 4 and 5, so $\frac{4}{9}$ is below a half; half of 8 is 4, so $\frac{5}{8}$ is above a half. $\mathbf{\frac{4}{9} < \frac{5}{8}}$

Try it: Which is greater, $\frac{7}{12}$ or $\frac{5}{9}$? (Both are just above a half, so go to a common denominator.)

check: $\mathbf{\frac{7}{12} > \frac{5}{9}}$

3. Ordering a whole set across representations

Method:

- **Sort by size class first.** Separate the numbers below one from those above one — often half the ordering is done for free.
- **Convert to one form.** Mixed numbers are usually easiest: turn every fraction greater than one into a mixed number.
- **Compare whole parts, then fraction parts.** Only when the whole parts tie do you need a common denominator for the fraction parts.
- **Write the answer in the original forms**, joined by $<$ or $>$, so the reader can see which given number went where.

Example: Order $\frac{5}{6}$, $1\frac{1}{4}$, $\frac{7}{8}$ from least to greatest.
Step 1. Size class settles part of it immediately: only one of these is above one, so it must be the largest and the other two need a real comparison.
Step 2. $\frac{5}{6}$ and $\frac{7}{8}$ are both below one, and over twenty-fourths they are $\frac{20}{24}$ and $\frac{21}{24}$. $\mathbf{\frac{5}{6} < \frac{7}{8} < 1\,\frac{1}{4}}$

Try it: Order $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{6}$ from least to greatest.

check: $\mathbf{\frac{2}{3} < \frac{3}{4} < \frac{5}{6}}$

(!) Watch out: a bigger denominator does *not* mean a bigger number — it means smaller pieces. And never compare numerators across different denominators; rewrite first.